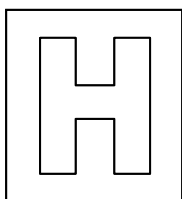


Candidate Name: _____

Class Adm No

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2013 Preliminary Examination II Pre-University 2

H1 Biology

8875/02

23 September 2013

2 hours

Additional Materials: Writing paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Index number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions on the question booklet.

Section B

Answer **one** question on writing paper.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question. At the end of the examination, fasten all your work securely together.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
Section B	
Total	

This question paper consists of 12 printed pages.

[Turn over

Section A

Answer **all** questions in this section.

1. A scientist carried out an investigation to determine the effect of a newly discovered compound Y on the rate of an enzyme catalysed reaction. The results of a control experiment without Y added and an experiment with Y added are shown in the graph in Fig. 1.1.

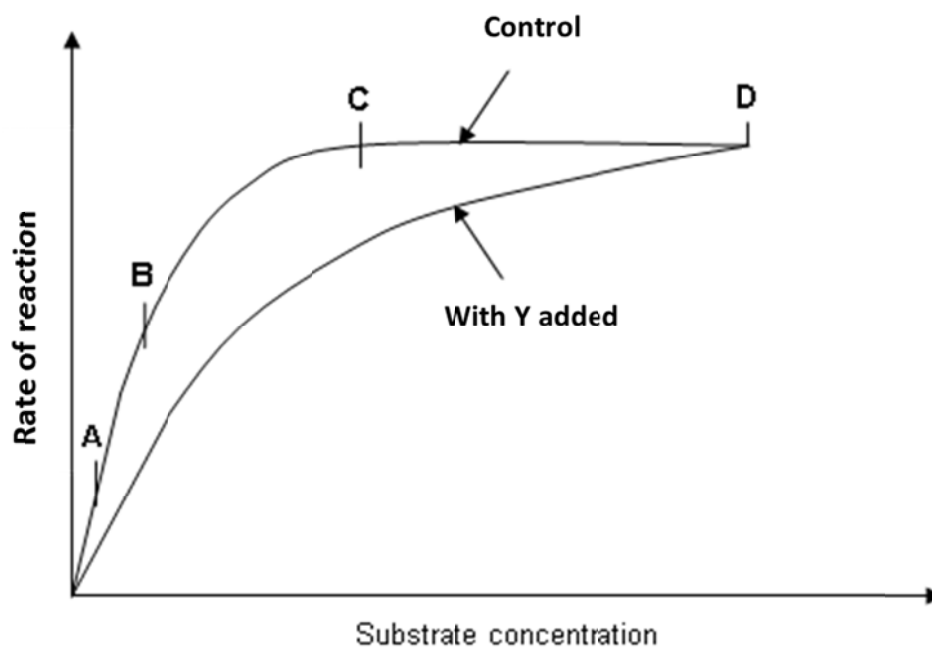


Fig. 1.1

- (a) With reference to Fig. 1.1, suggest the identity of Y and explain the reasons for the difference between the two graphs between points **A** and **B**.

rom

[3]

(b) Explain why the two graphs eventually reach the same maximum at point D.

- 1. At high substrate concentrations, there is a higher probability of substrate binding to active site than inhibitor;
- 2. At this point, presence of inhibitor is negligible and does not affect the maximum rate of reaction.

[2]

(c) Suggest two mechanisms by which an enzyme can lower activation energy of a reaction.

- 1. Organising substrates in proximity and correct orientation;
 - 2. Covalent catalysis by forming temporary covalent bond between enzyme and substrate;
 - 3. Bond strain, where crucial bonds in substrate are strained by enzyme;
 - 4. Providing favourable microenvironment for reaction to proceed.
- Any 2

[2]

[Total: 7]

2. FCCP (carbonyl cyanide-*p*-trifluoromethoxyphenylhydrazone) is a hydrophobic molecule that can cross the membrane freely as shown in Fig. 2.1. It is able to interconvert between the forms FCCP^- and FCCPH and can transport protons freely across the membrane.

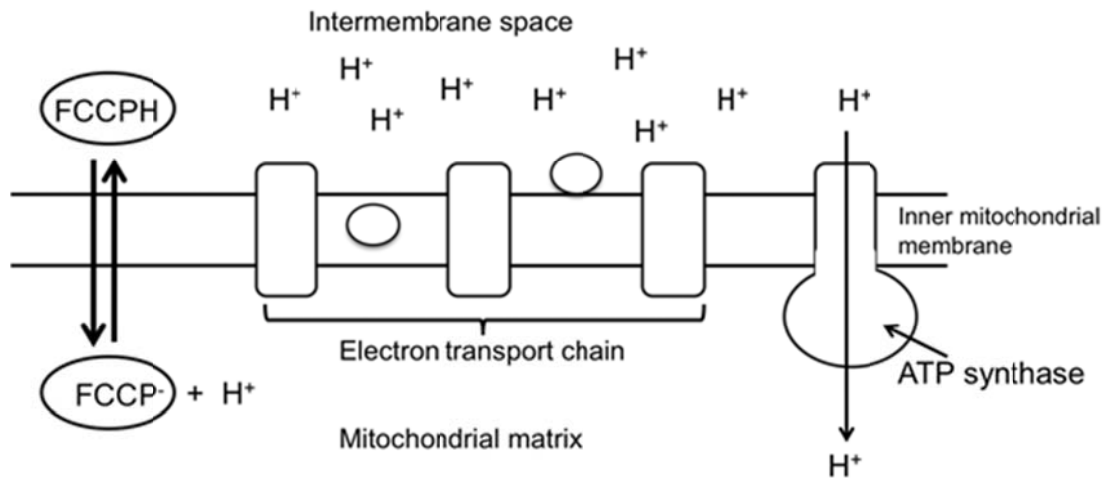


Fig. 2.1

A student added a small quantity of FCCP to cells isolated from rat liver. He observed that this addition resulted in an increase in oxygen consumption by the cells, however, ATP production decreased.

- (a) Account for the two observations made by the student upon addition of FCCP.

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..... [4]

Fluoroacetate is an inhibitor of aconitase, one of the enzymes involved in the Krebs cycle.

- (b) If fluoroacetate was added to cells instead of FCCP, suggest the effect of this addition on the process of oxidative phosphorylation.

1. Inhibition of Krebs cycle would reduce quantity of NADH/FADH₂ produced through dehydrogenation; R no production of NAD
2. Fewer NADH/FADH₂ to supply electrons and hydrogen ions to the electron transport chain;
3. Decrease in rate of oxidative phosphorylation. R stop oxidative phosphorylation

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..... [3]

- (c) Describe the role of NAD in aerobic respiration.

1. Hydrogen and electron carrier;
2. Required for reduction/dehydrogenation of compounds in glycolysis, link reaction and Krebs cycle;
3. Transports hydrogen and electrons to the electron transport chain in oxidative phosphorylation;
4. Transfers electrons to first electron carrier of the ETC;
5. Releases H⁺ ions to the matrix;
6. Recycled to NAD⁺ to allow continuity of aerobic respiration processes.

Any 4

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..... [4]

[Total: 11]

3. Rickets is a disease that often affects children, causing softening and weakening of bones. One of the causes of this disorder is caused by a mutation in an allele found on the X chromosome.

(a) Draw two genetic diagrams to show how:

- two parents who do not have the disease could have a son with rickets.
- a man with rickets and a woman without rickets could have a daughter with rickets.

Cross 1

Parental phenotype
Parental genotype

Normal
 $X^R X^r$

x
x

Normal
 $X^R Y$ [1]

Gametes

X^R

X^r

X^R

Y [1]

Punnett square

	X^R	X^r
X^R	$X^R X^R$	$X^R X^r$
Y	$X^R Y$	$X^r Y$

F1 genotype

$X^R X^R$

: $X^R X^r$

: $X^R Y$

: $X^r Y$

F1 phenotype

2 normal female

: 1 normal male

: 1 rickets male [1]

Cross 2

Parental phenotype
Parental genotype

Normal
 $X^R X^r$

x
x

Rickets
 $X^r Y$ [1]

Gametes

X^R

X^r

X^r

Y [1]

Punnett square

	X^R	X^r
X^r	$X^R X^r$	$X^r X^r$
Y	$X^R Y$	$X^r Y$

F1 genotype

$X^R X^r$

: $X^r X^r$

: $X^R Y$

: $X^r Y$

F1 phenotype

1 normal female

: 1 rickets female

: 1 normal male : 1

rickets male [1]

[6]

[Total: 6]

4. Fig. 4.1 shows two sections from an onion root tip undergoing mitosis.

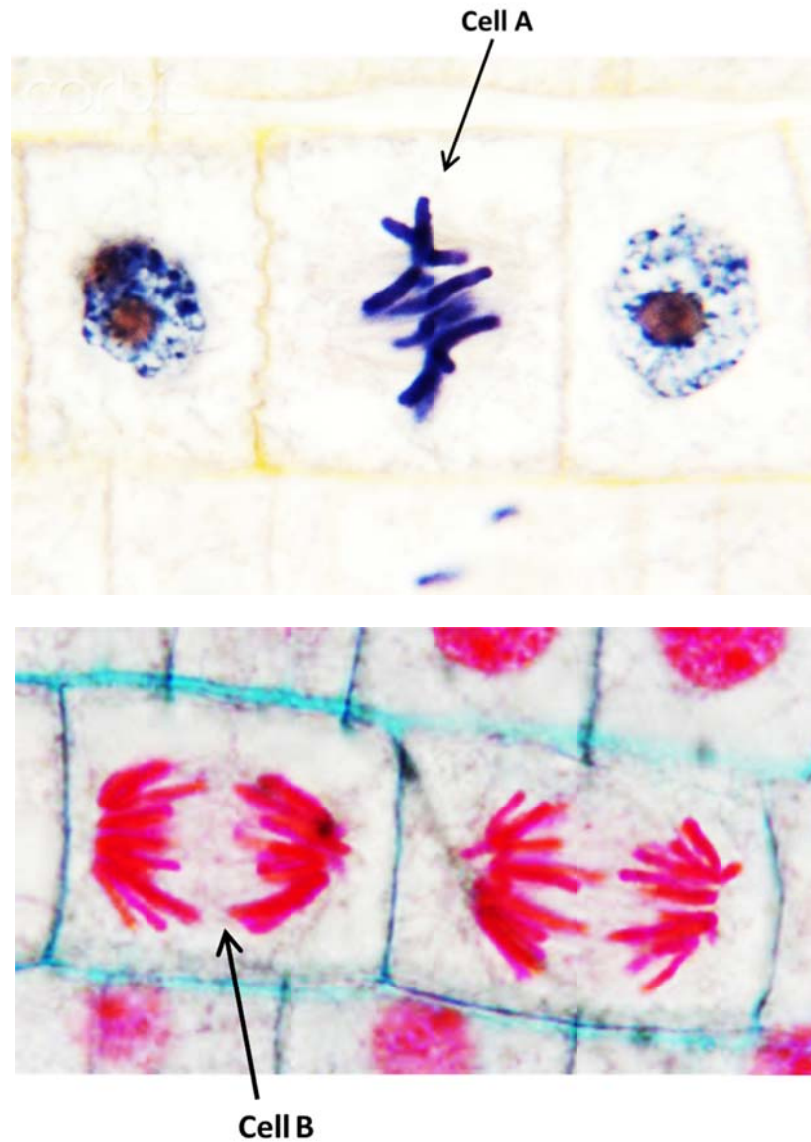


Fig. 4.1

- (a) State the stage of nuclear division that cells A and B are undergoing.

Cell A:		
Cell B:		 [2]

- (b) Describe what occurs in the cell during the two stages that you have identified in (a).

Metaphase

1. Chromosomes line up in a single row at the equator/metaphase plate;
2. Spindle fibres attach to kinetochore at centromeres of chromosomes;

Max 2

Anaphase

3. Spindle fibres shorten; R contract
4. Centromeres divide and sister chromatids separate;
5. Sister chromatids are pulled to opposite poles.

Max 2

.....
..... [4]

- (c) Explain the significance of mitosis in cell division.

- .. 1. Allows production of two genetically identical daughter cells;
- .. 2. Required for tissue regeneration during wear and tear;
- .. 3. For growth in multi-cellular organisms;
- .. 4. For asexual reproduction in single cellular organisms
- .. Any 2

..... [2]

[Total: 8]

5. A zombie outbreak is currently threatening the survival of humanity. Scientists have discovered that the cause of the zombie outbreak is a result of a 35 base pair deletion in a gene named *zomb*, coding for protein carrier Z. As a result of this mutation, a new restriction site for the restriction enzyme BamHI is introduced. Fig. 5.1 shows the restriction map for normal *zomb* and mutant *zomb*.

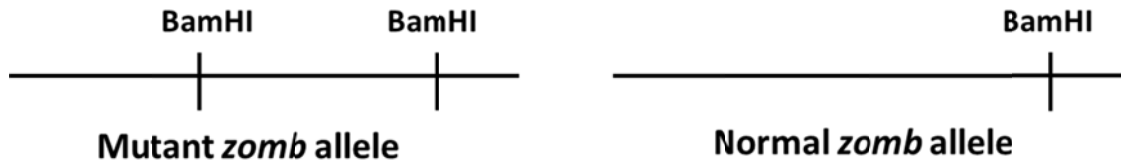
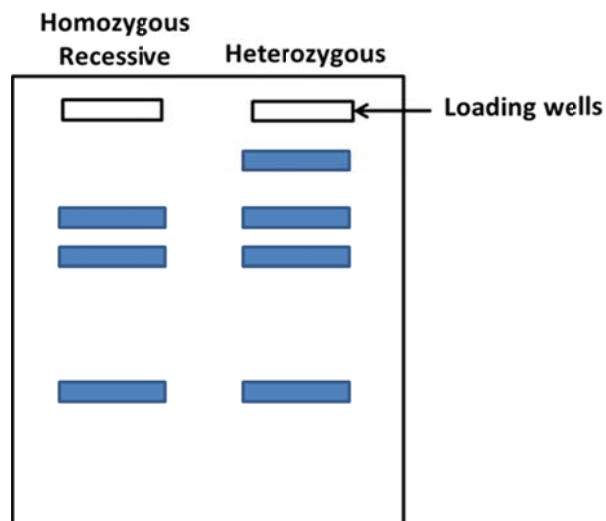


Fig. 5.1

The DNA samples of an individual who is homozygous recessive for the mutant *zomb* allele and an individual who is heterozygous for the *zomb* gene locus are subjected to restriction digestion by BamHI. The digested samples are then analysed via gel electrophoresis.

- (a) Draw the restriction pattern of these two individuals when their restriction fragments are analysed via gel electrophoresis.



[2]

- (b) Explain how restriction fragments are visualised during gel electrophoresis.



[2]

A bite from a zombie is usually sufficient to convert a normal individual into a zombie. However, a study done by the World Health Organisation (WHO) showed that the time lapse between the moment an individual is bitten to the time he/she is converted into a zombie varies greatly within a population. Based on these observations the WHO constructed a 5-point scale to express this variation. The scale is shown in Fig. 5.2. The variation observed was found to be a result of different allele variants of a particular gene X.

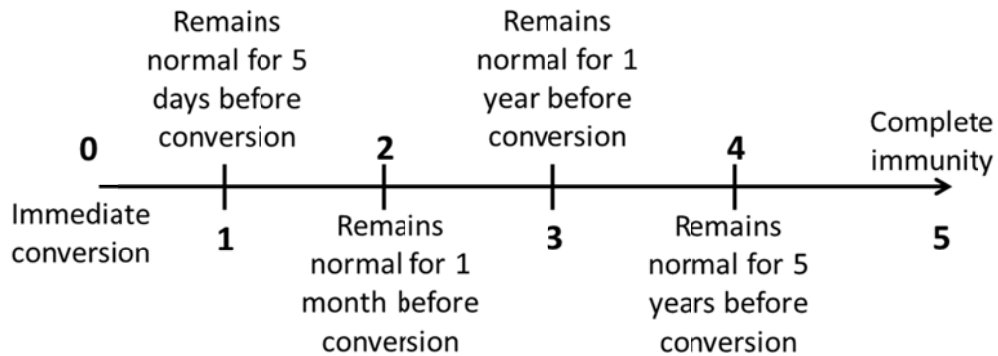


Fig. 5.2

- (c) State what type of natural selection would act on the population as a result of the zombie outbreak.

... .. [1]

- (d) With reference to the scenario given, use Darwin's theory of natural selection to explain how the allele frequencies of gene X in the population would change over time.

... [3]

[Total: 8]

Section B**Answer EITHER 6 OR 7.**

Write your answers on the separate answer paper provided.
 Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
 Your answers must be in continuous prose, where appropriate.
 Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

EITHER

6. (a) Using a named example, explain how the structure of a fibrous protein is related to its function. [8]

1. Clearly named example - Collagen;
2. Functions as a structural protein;
3. Made up of three polypeptide chains;
4. Each polypeptide chain is in helical conformation;
5. Cross-linking between three strands to form triple helix / tropocollagen
6. Many tropocollagen cross-linked via covalent bonds to form collagen fibre
7. 20-100 collagen fibres combine;
8. Contributes to tensile strength;
9. Rich in glycine and proline / Every third residue is glycine;
10. Allows close packing of three polypeptide chains;
11. Large proportion of hydrophobic amino acids;
12. And large molecular size;
13. Contributes to it being insoluble in water.

Note: Alternative examples e.g. keratin may be accepted, however, explanations must correspond to the example given.

- (b) Describe the pathway taken by insulin from the time it is synthesised till it is secreted from the cell. [6]

1. Insulin is synthesized as proinsulin by ribosomes attached to the RER;
2. Newly synthesised polypeptide enters lumen of RER;
3. Folding of polypeptide 3D structure in RER;
4. Buds off RER in transport vesicle and transported to Golgi apparatus;
5. Vesicle fuses with cis-face of Golgi;
6. Protein undergoes modification in Golgi to form mature insulin;
7. Packaged into secretory vesicles and bud off trans-face of Golgi;
8. Secretory vesicles travel from Golgi to cell surface membrane;
9. Vesicle membrane fuses with cell surface membrane to allow release of insulin via exocytosis.

(c) Discuss the functions of triglycerides in animals.

[6]

1. Form of energy storage;
2. Stores twice as much energy as carbohydrates of equal mass;
3. Because it has as higher proportion of hydrogen compared to carbohydrates;
4. Cushions vital organs such as kidneys against damage from impact;
5. Serves as thermal insulator under the skin;
6. Can be used for buoyancy for aquatic animals as it is less dense than water;
7. Oxidation of fat can produce metabolic water – useful for desert animals;
8. Can act as solvent for fat-soluble vitamins.

[Total: 20]

OR

7. (a) Using a named example, explain how a change in DNA sequence may affect the phenotype of an organism.

[4]

1. State example clearly i.e. Sickle cell anaemia
2. Change in DNA sequence will affect sequence of amino acids in the protein
3. ref. to A to T substitution mutation in α -haemoglobin gene resulting in change from valine to glutamic acid
4. Leading to change in phenotype - agglutination of haemoglobin S and red blood cells forming sickle shape

Alternative examples like cystic fibrosis and phenylketonuria are accepted – explanations must be relevant to example given.

(b) Discuss how gene or chromosomal mutations may result in cancer.

[8]

1. Gene mutations within tumour suppressor genes may lead to cancer;
2. Results in synthesis of non-functional tumour suppressor protein;
3. Leads to uncontrollable cell growth and division – causes formation of tumours;
4. Recessive mutation – requires 2 copies of mutant allele before there is a phenotypic effect
5. Gene mutations within proto-oncogenes may convert them to oncogenes;
6. Results in hyperactive version of proto-oncogene;
7. Overstimulation of cell growth and division – causes formation of tumours;
8. Chromosomal mutations such as duplication may lead to amplification of proto-oncogenes;
9. Translocation may lead to proto-oncogenes coming under control of a highly active promoter/control element;
10. Dominant mutation – only requires 1 copy of mutant allele for there to be a phenotypic effect.
11. Ref. disruption to cell cycle checkpoint
12. Ref. activation of telomerase

No marks if no reference to tumour suppressor and proto-oncogenes

- (c) Describe how the information on DNA may be used to synthesise a mature mRNA. [8]

1. Transcription of DNA to complementary pre-mRNA;
2. By RNA polymerase;
3. Transcription begins at promoter;
4. Pre-mRNA undergoes post-transcriptional modification to mature mRNA;
5. In nucleus prior to export to cytoplasm;
6. 5' capping, addition of modified guanine to 5' end of mRNA;
7. Splicing, excision of introns and splicing together of exons;
8. Ref. to alternative splicing;
9. 3' polyadenylation, addition of 3' poly(A) tail.
10. Correct sequence of post-transcriptional modification.

[Total: 20]

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